CS 253 longest-palindrome problem

You are given S, a string of length N. You get a subsequence by deleting zero or more letters from S. You want to find a longest subsequence of S which is a palindrome. That is, it equals its own reversal.

If there are multiple solutions, you prefer the leftmost subsequence. That is, you want a subsequence whose first symbol is as far left as possible in S. Among those solutions with the same first symbol, you want the second symbol to be as far left as possible, and so on.

Remark: for most of the test cases, N is at most 5000, and a well-written quadratic time solution should suffice. For the last test (worth 5%) N is 40000, but the number of deletions needed to get the subsequence is at most 10, so you could try to handle that as a special case (some faster heuristic method).

Input Format:

The first line is a single integer N.

The second line is a string S of length N.

Constraints

$1 \leq N \leq 5000$ (except N=40000 in the last test)

All characters of S are lower-case alphabetic letters ('a' to 'z', inclusive)

Output Format

The first line a single integer, the length of the subsequence.

The second line is the palindromic subsequence.

Sample Input 0

9

abcabcabc

Sample Output 0

5

abcba

Sample Input 1

10

aabbbaaaba

Sample Output 1

7

aabbbaa

Sample Input 2

10

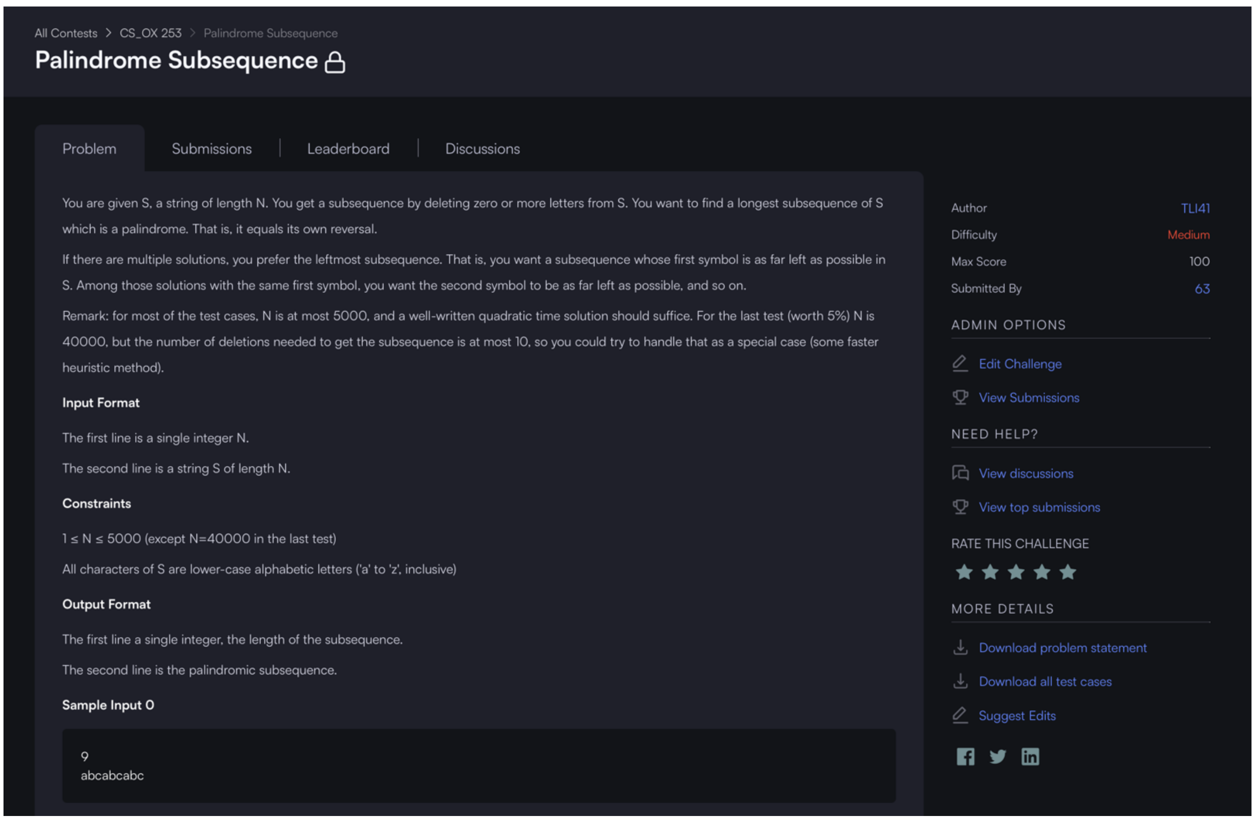
bababbaabb

Sample Output 2

8

baabbaab

Screenshot on Hankerrank:



All Contests > CS_OX 253 > Palindrome Subsequence

Palindrome Subsequence

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Sample Input 0

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Author	TLI41
Difficulty	Medium
Max Score	100
Submitted By	63

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